

Year 7

Year 8

IGCSE

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Homepage

Section 3.4: Magnetic Storage Media

In this section you will see the 4 different types of magnetic storage media.

- Fixed hard disk drive
- Portable hard disk drive
- Floppy disk
- Magnetic tape

For each we will discuss **features, uses, advantages/disadvantages**.

Key Concepts of this section:

- # Know the **4 different types** of magnetic storage media.
- # Understand the **uses** of each magnetic media.
- # Be able to **discuss advantages and disadvantages** of each magnetic media.

Examples of Magnetic Storage Media



Key Words:

Fixed Hard Disk Drive, Portable Hard Disk Drive, Floppy Disk, Magnetic Tape

The table below shows you some examples of magnetic storage media. Click each of the media to move to more information:

Fixed Hard Disk Drive



(Direct Access)

Portable Hard Disk Drive



(Direct Access)

Floppy Disk



(Direct Access)

Magnetic Tape



(Serial Access)

REMEMBER!

Magnetic storage media uses a magnetised surface area in order to hold bits of information.

NOTE:

Click the images to the left to move to more information about each one.

Breakdown of each output device:

The following sections will help summarise each of the 4 different magnetic storage media. For each one you will see:

- **Information about the media**
- **Uses of the media**
- **Advantages of the media**
- **Disadvantages of the media**

1 Fixed Hard Disk Drive

Information:

Used on **all computers** and are the **main method** of storing data. This is because they are capable of very fast **direct access** of files and have **very large storage** capacities.

The disc surface (Platter) is coated in a **magnetic film** which is where the data is stored.

These have **read/write heads** which allow data to be written to (saved) or read (used) from the disc.

The read/write heads (along with **direct data access**) allow for very fast data access speeds.

Hard Discs are used to store **operating systems, software applications** and all **other files**.

Typical storage spaces are **250 gigabytes up to 4 terabytes** (4,096 gigabytes).

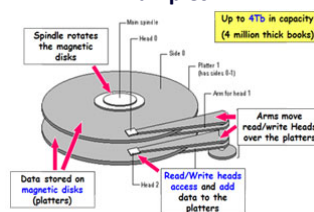
Uses of Fixed Hard Disk Drives:

- # Used to store the **operating system** and **applications software** (Windows, Word, Excel etc).
- # Used for storing **files** and **documents** (music and homework for example).
- # **Real-time systems** (robots, chemical plant control systems) and **online systems** (booking airline tickets etc) use hard disc drives to store data.

This is because **real-time / online systems** require data to be **processed very quickly**. Fixed hard disk drives have very fast data access speeds and so can handle this.

- # Used in **file-servers** on **computer networks** to store files (Like the student: drive in your school is used to store your work).

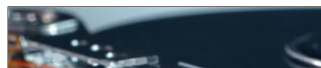
Examples:



A summary of a fixed hard disk drive.
(Click to zoom)

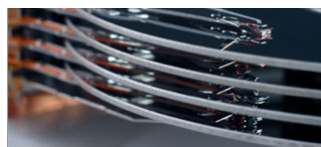


Fixed hard disk drives are designed to be 'fixed' into the computer and not removed.



Advantages/Disadvantages of Fixed Hard Disk Drives:

Advantages	Disadvantages
Fast read/write times (Saves data to disk and reads back from it quickly).	Can be damaged easily when computer is not shut down properly (Disc crash!).
Huge capacities (Can store thousands of movies and music files).	Not portable as they are designed to be 'fixed' into computer and not removed
Very easy to update/delete files	



Fixed hard disk drives store data on magnetically coated disks (known as platters).

2 Portable Hard Disk Drive

Information:

- # Works in a similar way to fixed hard discs but are **connected to the computer externally** via a Universal Serial Bus (**USB**) port.
- # They use **direct data access methods** which makes reading/writing of data very fast.
- # Portable Disc Drives are designed to be **transportable**.
- # They come with a **USB Cable** to allow for easy **attachment to other computers** which allows for easy backing up / sharing of files.
- # Because they are used outside of the computer they come with **protective casing** to **avoid physical damage** to the disc drive.
- # Typical storage spaces are **250 gigabytes up to 3 terabytes** (3,072 gigabytes).

Uses of Portable Hard Disk Drives:

- # Like fixed hard drives, they are used to **store large files** (such as video).
- # Can be used as **portable back-up systems** to avoid loss of data.
- # Used to **transfer data, files and software** between computers via USB connectivity.

Their **protective casing** makes them harder to damage whilst in transport.

Advantages/Disadvantages of Portable Hard Disk Drives:

Advantages	Disadvantages
Fast read/write times (direct data access).	Small size makes them easy to misplace or have stolen .
Huge capacities . (Thousands of movies and music files)	More expensive than other forms of storage
Small and light with protective casing. (Makes them very portable)	Protective casing will only protect against minor bumps. If the device is dropped it can still become damaged .
Designed to plug into almost any computer via USB ports.	Note: This is always a risk when transporting data externally!

3 Floppy Disk

Information:

- # Old method of **portable data storage** where information is held on a **thin plastic disc**.
 - # Data on a floppy disk is **read/written** by a **floppy disk drive** which rotates the disk.
 - # As the disc rotates, a **read/write head** is used to **add or read data**.
- The data is read/written using **direct data access** methods.

Even though floppy disks use direct data access, their **data transfer rates are very slow**. Good job they can only store 1.44MB!!

- # The disc of plastic is protected by a **retractable metal sleeve** and **hard plastic casing**. A **small notch** provides **write protection** (prevents accidentally copying over work).

Examples:

A summary of a portable hard disk drive. (Click to zoom)



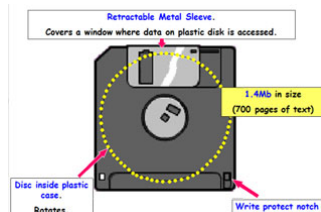
Portable hard disk drives are protected from damage by a tough outer casing.



Portable hard disk drives are small and light.



Portable hard disk drives connect to almost any other computer via usb..

Examples:

A summary of a floppy disk. (Click to zoom)



- # maximum **storage** on a floppy disc is only about **1.44 MB** (700 pages of text).

Floppy disks **do not have high enough capacities** to store **large modern files**. This has made them **obsolete** (useless) and computers no longer have floppy disk drives installed.

Uses of Floppy Disks:

- # Can still be used where **very small files** need to be **transferred/stored** (e.g. small word processed documents).
- # **Write protect facility** is useful to **prevent accidental overwriting of data**.

Advantages/Disadvantages of Floppy Disks:

Advantages	Disadvantages
Cost very little to buy.	Very low storage capacity when compared to other media (1.44 Mb).
Can be write protected easily which protects against accidentally copying over files.	Very few modern computers have floppy disc drives (device used to read the disk).
Very small, light and easy to carry (portable).	Floppy discs are very delicate and easy to damage (Not robust).
	Slow data transfer rate (Takes a long time to save/read from the disk).



The magnetic plastic disk is protected by plastic casing and a metal sleeve.



Floppy disks are read/written to using floppy disk drives.



Floppy disks are now doomed!

4 Magnetic Tape

Information:

- # This is one of the **oldest forms** of computer data storage.
- # Magnetic tape is a **thin strip of magnetic coated plastic** which is **wrapped onto a reel**.
- # Data is stored on the magnetic plastic in the form of **1's and 0's** (binary numbers).
- # Data is **written to and read from in sequence** (i.e. in order) which is also known as **'serial access'**.

Remember: Serial Access works a bit like a **video tape**. To access something in the middle of the tape you need to start at the beginning then fast-forward until you get to the part you need.

- # This type of storage media has a **very slow data access rate**.

Because it is so slow, It is **useless for real-time applications** (where what is stored is constantly being updated).

The slow access speeds makes it useless for holding data during real-time applications (e.g. alarm systems). This is because in real-time applications, stored data needs to be **constantly and quickly updated**.

- # Sony have developed magnetic tape capable of storing a huge **185TB** of data.

Uses of Magnetic Tape:

- # Used where **extremely large amounts of data** need to be **backed up**.
- # Used where **speed of reading/writing** of data is **NOT a priority**.

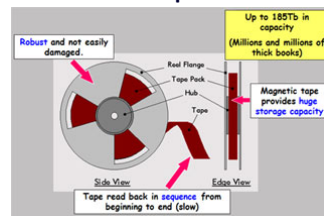
For Example: Magnetic tape is used to hold data in **batch processing applications** such as **clearing bank cheques** and **producing payslips**.

It can be used here because speed is not important. These jobs are usually left running at night when everyone has gone home.

- # Used for **backups of computer networks** in schools and business.

For Example: Your school files are frequently backed up on the school's network.

Examples:



A summary of magnetic tape.
(Click to zoom)



Magnetic tape is a magnetic coated plastic wrapped onto a reel.



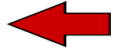
Magnetic tape uses serial (sequential) access to read/write data. It works much like a movie film reel.

Advantages/Disadvantages of Magnetic Tape:

Advantages	Disadvantages
Less expensive than the equivalent capacity hard disk drive.	Very slow data access/transfer (Reading data back from the tape is slow).
Very robust (Not easily damaged).	Needs another tape to update data
Very large storage capacities (Up to 185 terabytes or 185 trillion characters).	For example: original tape + tape with the changes = updated tape.

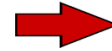


Magnetic tape is one of the oldest forms of computer data storage media and it can store vast amounts of data.



[Previous - Storage Media Introduction](#)

[Next - Optical Storage Devices](#)

**Links to Theory Units:**

[Section 1: Types and Components of Computer Systems](#)

[Section 2: Input and Output Devices](#)

[Section 3: Storage Devices and Media](#)

[Section 4: Computer Networks](#)

[Section 5: Data Types](#)

[Section 6: The Effects of Using ICT](#)

[Section 7: The Ways in which ICT is used](#)

[Section 7: Part 2](#)

[Section 8: Systems Analysis and Design](#)

Links to Practical Units:

[Section 9: Communication](#)

[Section 10: Document Production](#)

[Section 11: Data Manipulation](#)

[Section 12: Integration](#)

[Section 13: Output Data](#)

[Section 14: Data Analysis](#)

[Section 15: Web site Authoring](#)

[Section 16: Presentation Authoring](#)

[Back to top](#)